

SUBODH KUMAR

ML Engineer | AI Backend Developer | RAG & NLP Systems

Greater Noida, U.P., India | +91-7007456072 | subodh.24fd@gmail.com

LinkedIn: [linkedin.com/in/er-subodh-kumar](https://www.linkedin.com/in/er-subodh-kumar) | GitHub: github.com/ksubodh9

ML Engineer with 4.5+ years of backend engineering experience, specialising in production-ready AI systems, RAG pipelines, predictive analytics, and ML model integration. Strong foundation in Python, FastAPI, and scalable backend architecture — enabling end-to-end delivery of AI products from model development to deployment. Hands-on experience in fintech and AI domains with a focus on reliability, scalability, and real-world impact.

TECHNICAL SKILLS

ML & AI	Scikit-learn, XGBoost, Random Forest, Gradient Boosting, ARIMA/SARIMA, Predictive Analytics, Feature Engineering, Model Evaluation, Time-Series Forecasting, EDA
Generative AI / NLP	RAG Pipelines, FAISS, Vector Databases, Semantic Search, Embeddings, Prompt Engineering, Conversational AI, LLM API Integration
Backend & APIs	Python, FastAPI, PHP (Laravel), REST APIs, JWT Authentication, Swagger UI, API Design & Model Serving
Deployment & Tools	Docker, Git, Streamlit, AWS S3, Linux (CentOS), Windows Server
Databases	MySQL, MSSQL, MariaDB
Visualization	Matplotlib, Seaborn, Power BI (Basic)

PROFESSIONAL EXPERIENCE

Vayam Technologies Ltd.

Aug 2023 – Present

Product Engineer / ML Integration Engineer | MaddoxPay Fintech Platform

Applied AI / ML Integration

- Integrated predictive ML models into internal dashboards for anomaly detection and operational insights, improving fraud detection accuracy.
- Built and deployed backend APIs for ML model serving and prediction workflows with reliability-focused fallback handling.
- Supported AI-enabled decision workflows through predictive analytics and backend automation, reducing manual review overhead.

Backend Engineering & Fintech Systems

- Designed and maintained high-volume recharge and bill-payment systems handling 100K+ monthly transactions.
- Implemented automated payout systems (₹1M+ daily) using Axis Bank integrations — NEFT, RTGS, and IMPS.
- Optimised database schemas and backend workflows, improving API response performance by ~2x.
- Built analytics dashboards and reconciliation systems for operational monitoring and risk analysis.
- Implemented MFA, geofencing, biometric eKYC integrations, and fraud prevention workflows.

Flyde Digital Solutions

Jul 2022 – Aug 2023

Fullstack Engineer

- Delivered backend-focused full-stack applications with scalable API integrations (Razorpay, PhonePe, CC Avenue, Shiprocket, DTDC).
- Worked on AWS S3-based media storage systems and improved backend performance through query optimisation and modular development.

Saatirmind Software Services

May 2021 – Jul 2022

Laravel Backend Developer

- Developed modular web applications using Laravel and MySQL; worked on backend APIs, database optimisation, and scalable workflows.

PROJECTS

Smart AI Assistant — RAG-Powered Conversational System

Tech: *Python, FastAPI, FAISS, Vector DB, RAG, LLM APIs, Laravel*

- Designed and built an end-to-end RAG pipeline including data collection, embedding generation, FAISS vector indexing, and context-aware response generation.
- Implemented semantic similarity search for efficient knowledge retrieval; integrated bilingual conversational support and file attachment processing.
- Handled prompt orchestration and retrieval optimisation for scalable AI backend integration.

Career Guidance Engine API — AI-Powered Recommendation Platform

Tech: *FastAPI, Python, RAG, FAISS, Vector DB, NLP, JWT, Voice APIs*

- Built a personalised AI career guidance platform with custom RAG pipeline for context-aware responses and semantic search.
- Integrated FAISS-based vector retrieval and voice assistant APIs for voice-enabled conversational interactions.
- Designed scalable FastAPI backend with JWT authentication and modular architecture.

Industrial Equipment Monitoring & Predictive Maintenance System

Tech: *FastAPI, Python, Scikit-learn, Streamlit, Docker*

- Engineered an AI-powered predictive maintenance platform; built preprocessing and feature engineering pipelines for sensor-based industrial data.
- Implemented and compared Random Forest, XGBoost, Logistic Regression models for fault classification and failure detection.
- Developed FastAPI backend services and Streamlit dashboards for real-time monitoring; structured Docker-ready modular architecture.

Customer Churn Prediction Platform

Tech: *Python, Scikit-learn, XGBoost, Pandas, NumPy*

- Built an end-to-end churn prediction system; trained and evaluated Logistic Regression, Random Forest, Gradient Boosting, AdaBoost, and XGBoost models.
- Achieved 80.3% accuracy using Logistic Regression after comparative evaluation; built reusable ML pipelines for preprocessing and prediction.

Walmart Sales Forecasting System

Tech: *Python, Pandas, Scikit-learn, ARIMA/SARIMA, Matplotlib*

- Developed retail sales forecasting using regression and time-series techniques; implemented trend, seasonality, and forecasting analysis.

Multimodal Biometric Authentication System

Tech: *Python, OpenCV, Computer Vision*

- Built a biometric authentication system using OpenCV; implemented image preprocessing, feature extraction, and biometric verification workflows.

EDUCATION & CERTIFICATIONS

Executive PG Certification — Data Science & Artificial Intelligence Intellipaat	2025
Microsoft SQL Certification Intellipaat	2025
B.Tech — Information Technology Dr. A.P.J. Abdul Kalam Technical University, Lucknow Virtual Lab Development Program (Bootathon) — IIT Bombay, 2018	2020